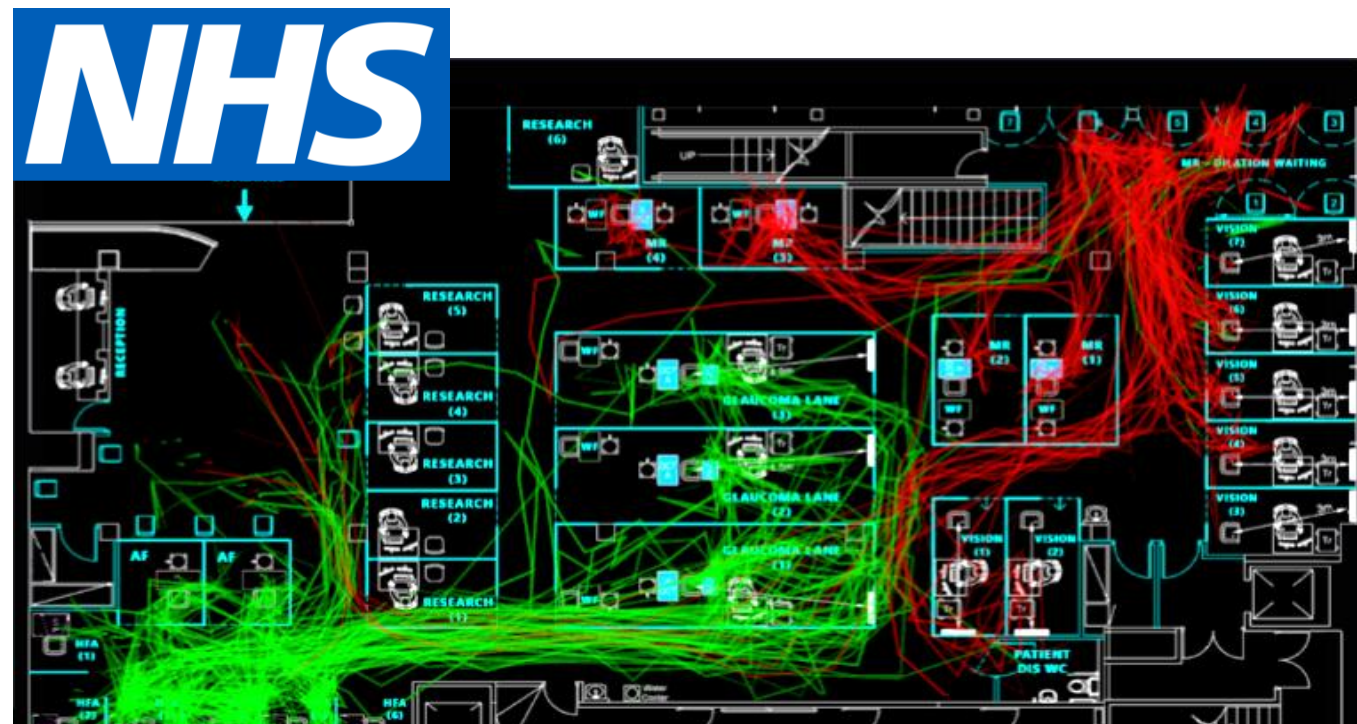


Why occupant tracking?

- Building optimisation & energy reduction.
- Shopper behaviour analysis.
- Optimising patient flow in healthcare clinics to reduce wait times.



Patient journeys being tracked and analysed in NHS clinics to reduce patient backlog [1]

The problem

- **Privacy Risks:** CCTV relies on centralised image storage, and tag-based systems (e.g., Ubisense [2]) are intrusive (50% participation refusal rate in NHS trials [1]).
- **Deployment Hurdles:** Relying on dedicated wired networks and centralised servers makes installation complex and limits scalability.

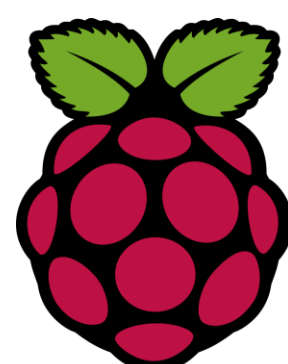
The goal

Developing a tracking system which is:

- **Privacy-Preserving.**
- **Scalable.**
- **Easy to deploy.**

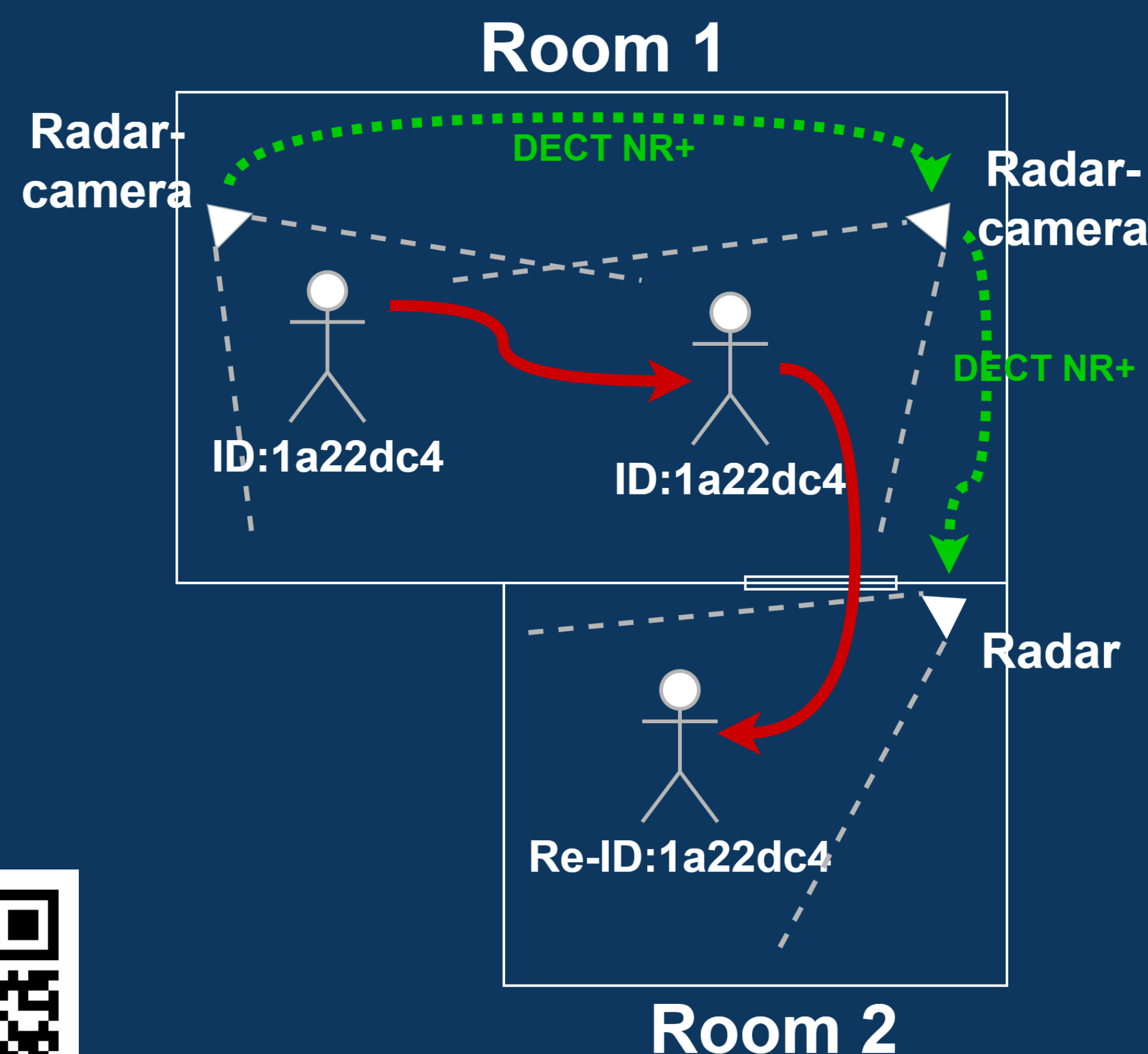
Proposed architecture

- **Edge Computing:** All data is processed locally on the nodes. Raw sensor data never leaves the sensor network.

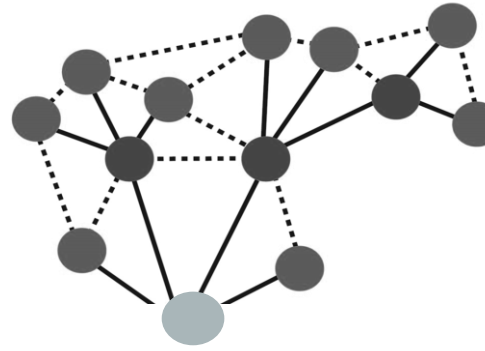


Project brief

Developing a privacy-first, scalable occupant tracking system using edge radar-camera fusion and Private 5G (DECT NR+)



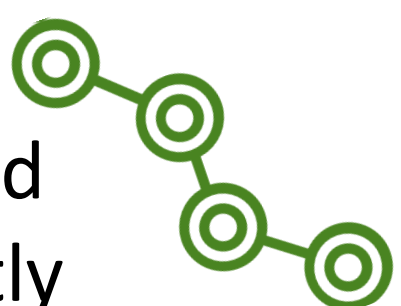
- **Mesh Networks:** Nodes self-coordinate on the 1.9 GHz DECT NR+ band (Non-cellular 5G), avoiding 2.4 GHz interference for lower latency and easier deployment.



- **Radar-Camera Fusion:** Fusing visual features with radar height and gait to create temporary, anonymous IDs, while leveraging radar depth and Doppler for continuous spatial tracking.



- **Cross-Scene Tracking:** Real-time feature handoffs build trajectories. Vectors are instantly purged, securely outputting only anonymised IDs and spatial coordinates.



Key Research Challenges

1. **Fisheye Distortion:** Maximising coverage with large-FoV cameras involves severe distortion. We utilise YOLO26-OBB to reliably bound these objects prior to fusion.
2. **Auto-Calibration:** Dynamic calibration requires (1) mapping local radar depth to 2D optical images, and (2) aligning spatial coordinates across overlapping node FoVs.
3. **DECT NR+ Mesh:** Implementing custom protocols on the new DECT NR+ PHY layer requires working with OFDM OTA synchronisation and dynamic routing.

Current Status: Early development stage.

References

- [1] <https://github.com/ucl-casa-ce/hercules-web>
[2] <https://ubisense.com/>